

STM32 32-bit MCU family

The fastest growing ARM® Cortex™-M based platform



Common core peripherals and architecture:

Communication peripherals: USART, SPI, I2C
Multiple general-purpose timers
Integrated reset and brown-out warning
Multiple DMA
2x watchdogs
Real-time clock
Integrated regulator PLL and clock circuit
Up to 3x 12-bit DAC
Up to 4x 12-bit ADC (up to 5 MSPS)
Main oscillator and 32 kHz oscillator
Low-speed and high-speed internal RC oscillators
-40 to +85 °C and up to 105 °C operating temperature range
Low voltage 2.0 to 3.6 V or 1.65/1.7 to 3.6 V (depending on series)
Temperature sensor

STM32 F4 series - High performance with DSP (STM32F401/405/415/407/417/427/437/429/439)

Up to 180 MHz Cortex-M4 DSP/FPU	Up to 2-Mbyte Flash	Up to 256-Kbyte SRAM	2x USB 2.0 OTG FS/HSG	3-phase MC timer	2x CAN 2.0B	SDIO 2x PS audio Camera IF	Ethernet IEEE 1588	LCD-TFT SDRAM I/F	
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STM32 F3 series - Mixed-signal with DSP (STM32F302/303/313/373/383)

72 MHz Cortex-M4 with DSP and FPU	Up to 256-Kbyte Flash	Up to 48-Kbyte SRAM & COM-SRAM	USB 2.0 FS	3x 3-phase MC timer (144 MHz)	CAN 2.0B	Up to 7x comparator	3x 16-bit 12.5 ADC	4x PGA	
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STM32 F2 series - High performance (STM32F205/215/207/217)

120 MHz Cortex-M3 CPU	Up to 1-Mbyte Flash	Up to 128-Kbyte SRAM	2x USB 2.0 OTG FS/HSG	3-phase MC timer	2x CAN 2.0B	SDIO 2x PS audio Camera IF	Ethernet IEEE 1588	Crypto	
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STM32 F1 series - Mainstream - 5 product lines (STM32F100/101/102/103 and 105/107)

Up to 72 MHz Cortex-M3 CPU	Up to 1-Mbyte Flash	Up to 96-Kbyte SRAM	USB 2.0 OTG FS	3-phase MC timer	Up to 2x CAN 2.0B	SDIO 2x PS audio	Ethernet IEEE 1588	
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STM32 F0 series - Entry level (STM32F030/50/051)

48 MHz Cortex-M0 CPU	Up to 64-Kbyte Flash	Up to 8-Kbyte SRAM	3-phase MC timer	Comparator	DEC	
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STM32 L1 series - Ultra-low-power (STM32L100/151/152/162)

32 MHz Cortex-M3 CPU	Up to 384-Kbyte Flash	Up to 48-Kbyte SRAM	USB FS device	Up to 12-Kbyte EEPROM	LCD 8x40 4x44	Op-amps comparator	BOR MSI Vscal	AES 128-bit	
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THE CHOICE IS YOURS WITHIN THE STM32 PLATFORMS, THE LEADING CORTEX™-M MCU FAMILY

- Comprehensive portfolio
- 16-Kbyte to 2-Mbyte Flash
- 20 pins to 216 pins
- Six series from the entry-level STM32 F0 to the high-end STM32 F4
- More than 450 pin-to-pin compatible part numbers
- Continued evolution of portfolio with new devices every year
- Perfect choice for a platform decision

START TODAY WITH STM32-DISCOVERY KITS

Discovery kits are the cheapest and quickest way to discover the STM32 family. These quick-start evaluation boards embed an ST-LINK or ST-LINK/V2 debug probe and are supported by IDE from Atollic, Keil, IAR and TASKING.



www.st.com

For more information, you can visit www.st.com/mcu or contact distributors

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